

Sex differences in whole-brain dynamics of the *Drosophila* CNS connectome

A matched leaky integrate-and-fire comparison with degree-preserving, sign-shuffle, and ablation-null controls

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Abstract

The completion of synapse-resolution connectomes for both the female (BANC) and male (MCNS) adult *Drosophila melanogaster* central nervous system enables a direct, whole-brain comparison of sex-specific wiring through dynamics. We embed both connectomes in an identical leaky integrate-and-fire model (weights from synapse counts, signs from neurotransmitter identity, matched 1% Poisson drive) and quantify recruitment and firing rates under a single-source-of-truth protocol: headline real-network conditions are run under three independent (weight-seed, simulation-seed) pairs, null networks under independent build seeds with dynamics re-seeded across the three simulation seeds (130 unique ledger records), and headline recruitment ratios carry bootstrap confidence intervals. Under lognormal weights matched to the literature's fitted log-spread ($\text{SD}(\log w) \approx 1.83$, obtained at $\sigma = 1.6$), the male recruits $79.0 \pm 0.2\%$ of neurons in a 2 s window versus $45.0 \pm 0.5\%$ in the female, a $1.75\times$ ratio (95% CI 1.74–1.78), with branching ratios $m = 0.980\text{--}1.004$ across the 130 unique ledger records. A degree-preserving double-edge-swap null decomposes the log-scale gap at this saturated anchor: **58% of the sex difference is destroyed by rewiring (specific wiring), 42% survives on degree/density structure alone**, and rewiring raises both networks' recruitment (female 45→72%, male 79→91%), so specific wiring reads as net-suppressive at this operating point; the split and its sign are operating-point-dependent and appear to invert near threshold (§3.3). The recruitment ratio is gain-dependent: $26\times$ (21.9–33.7) under linear weights, $14.5\text{--}17\times$ (mean-weight-matched lognormal sweep), $1.75\times$ at the literature-matched anchor, reflecting where on a shared recruitment curve each sex sits: the male reaches equivalent recruitment at $\sim 3\text{--}3.5\times$ lower synaptic weight. Higher-

order metrics that scale with firing rate (transfer entropy, participation ratio, CV of ISI) are shown by direct test to be highly susceptible to artifactual divergence between networks of different activity levels (§3.8). Silencing dimorphic neurons (2.4% of male, 2.2% of female cells) reduces male recruitment by 8.4% relative, versus 6.3% for size-matched random ablation, a modest, distributed effect with its proper null. We release the dual-seed protocol as a reproducibility standard for connectome-dynamics comparisons.

1. Introduction

The connectome specifies the synaptic paths by which neurons can influence one another, but not the dynamics that emerge from those paths (Pospisil et al., 2024). With synapse-resolution connectomes now available for the adult female CNS (BANC; 169,078 neurons) and male CNS (MCNS; 163,511 neurons), one can ask a question that brain-only comparisons cannot answer cleanly: **when two whole nervous systems that differ in sex-specific wiring are embedded in the identical dynamical model, do they produce measurably different dynamics, and is any difference attributable to specific wiring, to degree/density structure, or to localized sex-specific circuits?**

We answer this with an intervention-first design. Rather than comparing summary statistics of spontaneous activity and post-hoc explaining them, we run matched perturbations on both networks:

1. **Weight-rule manipulations** (linear, log1p, sqrt, lognormal with matched mean or matched spread) to test whether the sex difference survives defensible weight rules;
2. **Degree-preserving double-edge-swap (DES) rewiring** of each sex's own connectome, which preserves the exact in- and out-degree sequences while destroying specific wiring, isolating the contribution of *which* edges exist from *how many* each neuron has;
3. **Sign-shuffling** to test whether excitatory/inhibitory structure is net-suppressive or net-amplifying;
4. **Size-matched random ablation** as the null for dimorphic-neuron silencing.

All experiments flow into a single append-only results ledger (`ssot_results.jsonl`); every number in this paper is traceable to that ledger, to companion probe ledgers under `results/crit_probe/`, to archived spike-train artifacts, or to the shipped connectome edge lists and their sign annotations; headline real-

network conditions are sampled over three independent (weight, simulation) seed-pairs and null conditions over independent build seeds with dynamics re-seeded across the three simulation seeds.

A companion methodological result shapes the paper: metrics that scale with firing rate (transfer entropy, participation ratio, CV of ISI) diverge artifactually between networks of different activity levels. Section 3.8 quantifies these rate-confounding vulnerabilities and the controls required to neutralize them; this is a caution that applies to any connectome-dynamics comparison in which the networks under study differ in overall activity.

2. Methods

2.1 Data

Dataset	Sex	Scope	Neurons	Edges	Synapses	Mean count	SD(log count)	E/I edges
BANC v888	Female	Whole CNS	169,078	13,379,740	41.5M	3.10	0.825	75/25
MCNS v1.0	Male	Whole CNS	163,511	24,410,950	117.5M	4.81	0.983	80/20

BANC and MCNS are the primary comparison (both whole-CNS; BANC is described in Bates et al., 2025, and MCNS in Berg et al., 2026, extending the whole-brain wiring-diagram approach of Dorkenwald et al., 2024; global network statistics are catalogued in Lin et al., 2024). All data are publicly available (Janelia FlyEM, FlyWire Codex); no authentication was required. The male flat connectome was filtered to annotated neurons; neurotransmitter consensus calls map to signs (see §2.2).

2.2 Normalization

Each connectome was converted to a common edge list (`pre_idx`, `post_idx`, `count`, `sign`), where `count` is the synapse count and `sign` $\in \{+1, -1\}$ is derived from neurotransmitter identity:

- **Excitatory (+1):** acetylcholine, glutamate, dopamine, octopamine, tyramine
- **Inhibitory (−1):** GABA, histamine, serotonin, glycine
- **Unclear:** dropped (documented limitation; affects both sexes similarly)

Amine signs follow the ionotropic convention (dopamine, octopamine, and tyramine +1; serotonin −1); most amine receptors in *Drosophila* act through metabotropic pathways, so

this is a coarse modeling choice, and the sign-structure (§3.7) and E/I-confound (§3.3) conclusions are conditional on it (see Limitations).

2.3 Model

Both connectomes were embedded in an identical LIF model (Brian2, Cython codegen):

```
dv/dt = (v_rest - v)/τ (unless refractory); threshold v > v_th;
reset v = v_rest
```

with $v_{\text{rest}} = -65$ mV, $v_{\text{th}} = -50$ mV, $\tau = 10$ ms, refractory period 2.2 ms, simulated for 2000 ms. A uniformly random 1% of neurons (1,635 male / 1,690 female) received independent Poisson drive at 100 Hz, each event depolarizing the target by 5 mV. Synapses were $w = \text{sign} \cdot |w| \cdot \text{mV}$ with delays omitted (all-to-all chemical capture at 1 ms resolution is out of scope; see Limitations).

2.4 Weight transforms and the σ parameterization

Four rules map counts to weight magnitudes:

- **linear:** $|w| = \text{count} \cdot w_{\text{syn}}$
- **log1p:** $|w| = \log(1 + \text{count}) \cdot w_{\text{syn}}$
- **sqr:** $|w| = \sqrt{\text{count}} \cdot w_{\text{syn}}$
- **lognormal:** $|w| = \exp(N(\ln \max(\text{count}, 1), \sigma)) \cdot w_{\text{syn}}$ (drawn per edge from `weight_seed`)

For the lognormal rule, $\text{SD}(\log |w|) = \sqrt{(\text{SD}(\log \text{count})^2 + \sigma^2)}$. The recent connectome-constrained fitting literature (bioRxiv 2026.08.21.745055) reports fitted weight log-spread $\text{SD}(\log w) = 1.83$ (truncated-lognormal MLE). Matching that spread requires $\sigma = \sqrt{(1.83^2 - 0.983^2)} \approx 1.54$ for the male and $\sigma = \sqrt{(1.83^2 - 0.825^2)} \approx 1.64$ for the female; we adopt **$\sigma = 1.6$** as the literature-matched condition. Note $\sigma = 1.83$ itself would *overshoot* the fitted spread by inflating it with the count dispersion; the distinction between total log-spread and draw dispersion matters operationally. A 3-seed probe at $\sigma = 1.83$ ($\text{SD}(\log |w|) \approx 2.0$) confirms the overshoot reading: the ratio compresses further, to $1.47 \times [1.44, 1.50]$ (male $80.93 \pm 0.13\%$, female $55.17 \pm 1.05\%$), so the anchor choice is not innocuous; we retain $\sigma = 1.6$.

Two anchor regimes are reported side by side:

- **Spread-matched anchor:** $\sigma = 1.6$ at raw $w_{\text{syn}} = 0.1$. Mean $|w|$ exceeds the linear baseline (1.12 mV female, 1.73 mV male vs 0.31 mV), placing both networks in a high-recruitment regime.
- **Mean-matched sweep:** $w_{\text{syn}}(\sigma) = 0.1 \cdot \exp(-\sigma^2/2)$, which holds $E[|w|] = E[\text{count}] \cdot 0.1$ fixed while varying only

dispersion ($\sigma = 0.5 \rightarrow 0.08825$, $1.0 \rightarrow 0.06065$, $1.6 \rightarrow 0.0278$). This isolates the effect of weight *dispersion* from weight *gain*, a confound present in naive σ sweeps, where $E[|w|]$ grows as $\exp(\sigma^2/2)$.

2.5 Null models

1. **Degree-preserving double-edge swap (DES).** Starting from each sex's real edge set (64-bit keyed (pre, post) pairs), we perform $Q \cdot |E|$ attempted swaps ($Q = 10$; $\sim 134\text{M}$ attempts female, $\sim 244\text{M}$ male per seed) of random edge pairs, reassigning targets when the resulting edge set is edge-set-identical-free (rejection sampling). In- and out-degree sequences are preserved *exactly*, verified identical post-swap for every seed. Weights travel with their source edge. Acceptance rates: 99.1% (female), 98.7% (male). Three seeds per sex.
2. **Sign-shuffle.** Fisher-Yates permutation of the sign vector across edges, preserving the excitatory/inhibitory counts and all wiring.
3. **Size-matched random ablation.** All edges touching k uniformly random neurons are removed, $k = 3,900$ (male) / $3,803$ (female), exactly the dimorphic-neuron counts used in §2.7. Five build seeds per sex, each simulated under the three simulation seeds (15 runs per sex).

2.6 Single-source-of-truth (SSOT) protocol and statistics

Every simulation writes one JSON record to an append-only ledger keyed by (tag, source, weight_seed, sim_seed). Headline real-network conditions are sampled over three seed-pairs $(\text{weight_seed}, \text{sim_seed}) \in \{(0,42), (1,43), (2,44)\}$; four low- w_{syn} ladder cells (male w_{syn} 0.01, 0.03; female 0.02, 0.04) are single-seed (Table 5). Null networks carry their build seed as weight_seed (three DES seeds per sex; five ablation seeds per sex) and are simulated under all three simulation seeds: DES nulls are $3 \text{ build} \times 3 \text{ dynamics seeds}$ ($n = 9$ per sex), ablation nulls $5 \text{ build} \times 3 \text{ dynamics seeds}$ ($n = 15$ per sex). Cross-seed SD is small for the DES nulls (within-build ≤ 0.066 pp) but materially larger for the ablation nulls (within-build up to 2.65 pp male, 0.28 pp female), so ablation nulls are reported with their full spread. DES builds for all six networks (3 seeds $\times$ 2 sexes) were completed and degree-verified before their simulations were queued.

We report mean $\pm$ SD across seed-pairs (SD with ddof = 1) and two-sided 95% CIs on male/female ratios via 10,000-resample bootstrap over seed-pairs. With three seed-pairs, the bootstrap CI is a within-cell spread indicator rather than a resolved sampling distribution; narrow intervals reflect small within-cell SDs.

Aggregation deduplicates the ledger (130 unique records, 133 lines) by key with last-timestamp-wins. Recruitment is the fraction of neurons firing ≥ 1 spike in 2000 ms; population rate is spikes/s averaged over all neurons ("rate_all").

2.7 Metrics

- **Recruitment (% fired):** primary readout (robust to rate confounding by construction).
- **Population rate (Hz, all neurons) and per-active-neuron rate.**
- **Branching ratio m :** mean descendants per ancestor in 1 ms bins. At the drive densities used here (mean spikes per 1 ms bin $\lambda \geq 7.17$), the mean-ratio estimator is bounded above by $1 - \exp(-\lambda)$ and therefore saturates near 1 in every condition regardless of regime (§3.5).
- **CV of ISI:** reported but interpreted as rate-dependent (§3.8).
- **Transfer entropy, participation ratio:** rate-dependent; see §3.8 and Limitations.

2.8 Dimorphic-neuron identification and silencing

Sexually dimorphic neurons were identified from annotation flags (male: dimorphism, fruDs_x; female: sexually_dimorphic); 2.4% of male cells (3,900) and 2.2% of female cells (3,803). Silencing removes all edges touching dimorphic neurons (both sexes' own dimorphic sets). The size-matched random-ablation null (§2.5) provides the control.

3. Results

3.1 The recruitment gap exists under linear and lognormal weights, and vanishes under compression

Under linear weights at the reference drive ($w_{\text{syn}} = 0.1$), the male recruits $35.20 \pm 0.68\%$ of neurons versus $1.35 \pm 0.29\%$ in the female, a **26.1 $\times$ ratio (95% CI 21.9-33.7)** (Table 1). Under lognormal weights at matched mean weight, the ratio is 14.5–17 $\times$ across σ (Table 4). At the literature-matched spread anchor, both networks recruit heavily and the ratio compresses to 1.75 $\times$ (§3.2). The direction of the gap is invariant across every condition tested here (43/43 matched male $\geq$ female pairs in the ledger); its magnitude depends on where each sex sits on the recruitment curve.

Table 1. Baseline dynamics (2000 ms, linear weights, $w_{\text{syn}} = 0.1$; mean $\pm$ SD over 3 seed-pairs).

Metric	Female BANC	Male MCNS
Recruitment (% fired)	1.35 ± 0.29	35.20 ± 0.68
Population rate (Hz, all neurons)	0.07 ± 0.04	15.50 ± 3.85
Branching ratio m	0.999 ± 0.000	1.001 ± 0.002
CV of ISI	1.04 ± 0.22	2.74 ± 0.47

Under log1p and sqrt compression both sexes collapse to the $1.00 \pm 0.00\%$ drive floor (only directly driven neurons fire): the network sits at its drive floor and the sex gap vanishes. The male's excess activity is therefore carried by high-count connections; compress them and the gap disappears; this motivates the lognormal analysis, which allows strong edges without collapsing them.

3.2 Literature-matched lognormal weights: $1.75\times$ recruitment gap

At $\sigma = 1.6$, $w_{\text{syn}} = 0.1$ ($\text{SD}(\log|w|) \approx 1.83$, matching the fitted literature spread):

Table 2. $\sigma = 1.6$ anchor ($w_{\text{syn}} = 0.1$; mean $\pm$ SD over 3 seed-pairs).

Metric	Female BANC	Male MCNS	Ratio (M/F)
Recruitment (% fired)	45.04 ± 0.49	79.03 ± 0.16	$1.75\times$ [1.74, 1.78]
Population rate (Hz)	69.48 ± 16.40	183.21 ± 1.89	$2.64\times$
Per-active-neuron rate (Hz)	154.1 ± 35.4	231.8 ± 2.8	$1.50\times$
Branching ratio m	1.000 ± 0.000	0.998 ± 0.001	n/a

The male recruits $1.75\times$ more of the network, fires its active neurons $\sim 1.5\times$ faster, and integrates to a $2.6\times$ higher population rate. Note the female's population rate varies widely across seeds ($\text{SD} \pm 16.4$ Hz on 69 Hz, $n = 3$): at this operating point the female sits near an ignition threshold and individual seeds differ in how much of the network catches (see also §3.7). Recruitment, not rate, is the primary readout throughout.

3.3 Degree-preserving decomposition at the saturated anchor: 58% wiring, 42% degree structure

DES rewiring preserves each network's exact degree sequences while randomizing which edges exist. If the sex gap were a property of degree/density structure alone, rewiring would preserve it; if a property of specific wiring, rewiring would destroy it. The result is a decomposition:

Table 3. Real vs DES-rewired networks ($\sigma = 1.6$, $w_{\text{syn}} = 0.1$; DES: mean $\pm$ SD over 3 build $\times$ 3 simulation seeds, $n = 9$ per sex).

Network	Recruitment real	Recruitment DES	log-gap destroyed
Female BANC	45.04 $\pm$ 0.49%	72.11 $\pm$ 0.21%	0.471 log units
Male MCNS	79.03 $\pm$ 0.16%	91.46 $\pm$ 0.03%	0.146 log units

- Real log-gap: $\ln(79.03/45.04) = \mathbf{0.562}$
- DES log-gap: $\ln(91.46/72.11) = \mathbf{0.238}$
- **Specific wiring accounts for $(0.562 - 0.238)/0.562 = 58\%$ of the sex gap; degree/density structure accounts for 42%.**
- Rewired ratio: $1.27 \times [1.27, 1.27]$.

Two conclusions follow at this operating point. First, **two-fifths of the gap survives on degree/density alone (42%)**, so the difference is not purely a wiring property. Second, *both* brains' specific wiring reads as net-suppressive relative to their own degree structure: rewiring makes both networks more active (female +27 points, male +12 points). The female's wiring damps far harder per unit of degree (0.471 log units) than the male's (0.146). Three caveats bound this decomposition. The share is a log-ratio-scale quantity with seed uncertainty: bootstrapping over seed-pairs and DES builds gives 57.7% [56.9, 58.5] (10,000 resamples), while on the linear percentage-point scale the split reads 43% destroyed / 57% survived, so the 58/42 headline is a multiplicative statement, not a scale-free one. It is measured at the saturated anchor only: re-simulating the same DES edge sets across w_{syn} shows the specific-wiring share sliding from -233% ($w_{\text{syn}} = 0.04$) through $+30\%$ (0.06) to $+58\%$ (0.10), with the sign inverting near threshold (Table 3b). And DES rewiring does not only permute which edges exist, it also remixes each network's excitatory/inhibitory composition, which §3.7 shows is itself suppressively organized; wiring specificity and E/I arrangement are confounded in this null. The female's suppressive-wiring story should therefore be read as an anchor-specific, scale-dependent, E/I-confounded observation, not a general property.

Table 3b. The decomposition is operating-point-dependent (lognormal $\sigma = 1.6$; real and DES networks re-simulated across w_{syn} ; single seed-pair per point, anchor row 3-seed/9-record; the female $w_{\text{syn}} = 0.06$ cell is a single-seed reading (the 3-seed ladder row is 28.08 ± 0.22 , Table 5)).

w_{syn}	Real F	Real M	DES F	DES M	Ratio real	Ratio DES	Specific-wiring share
0.04	16.57	56.51	1.44	86.07	3.41	59.7	-233%
0.05	23.08	64.97	2.12	88.06	2.81	41.5	-260%
0.06	27.83	70.69	46.15	88.27	2.54	1.91	$+30\%$
0.07	33.07	74.17	61.12	89.40	2.24	1.46	$+53\%$
0.08	37.80	75.82	67.16	90.25	2.01	1.34	$+58\%$

w_{syn}	Real F	Real M	DES F	DES M	Ratio real	Ratio DES	Specific- wiring share
0.10 (anchor)	45.04	79.03	72.11	91.46	1.75	1.27	+58%

The DES female sits on a knife-edge transition (1.44% at $w = 0.04$ rising to 46.15% at $w = 0.06$, a $33\times$ jump for a $1.5\times$ weight change), which drives the sign inversion near threshold; the male is monotone and less fragile. The extreme negative shares at $w_{\text{syn}} = 0.04\text{--}0.05$ (-233% , -260%) are single-seed readings on this knife-edge: the sign inversion is plausible but indicative only, not seed-replicated.

3.4 Dispersion versus gain: the ratio and the decomposition are regime-dependent

Because $E[|w|]$ grows as $\exp(\sigma^2/2)$ under a raw sweep, σ sweeps confound weight dispersion with mean gain. The mean-matched sweep removes the confound:

Table 4. Weight-rule sensitivity (2000 ms; mean $\pm$ SD over 3 seed-pairs; ratios with bootstrap 95% CI).

Rule	Female	Male	Ratio
linear, $w_{\text{syn}} = 0.1$	$1.35 \pm 0.29\%$	$35.20 \pm 0.68\%$	$26.1\times [21.9, 33.7]$
lognormal $\sigma=0.5$, mean-matched	$2.15 \pm 0.44\%$	$36.75 \pm 1.00\%$	$17.1\times [14.2, 21.4]$
lognormal $\sigma=1.0$, mean-matched	$2.32 \pm 0.11\%$	$38.32 \pm 0.59\%$	$16.5\times [15.8, 17.2]$
lognormal $\sigma=1.6$, mean-matched	$2.81 \pm 0.33\%$	$40.93 \pm 0.35\%$	$14.5\times [13.4, 16.7]$
lognormal $\sigma=0.5$, raw $w_{\text{syn}}=0.1$	$3.98 \pm 2.07\%$	$38.12 \pm 2.97\%$	$9.6\times [6.3, 18.6]$
lognormal $\sigma=1.6$, raw $w_{\text{syn}}=0.1$	$45.04 \pm 0.49\%$	$79.03 \pm 0.16\%$	$1.75\times [1.74, 1.78]$
log1p	$1.00 \pm 0.00\%$	$1.00 \pm 0.00\%$	$1.0\times$ (drive floor)
sqrt	$1.00 \pm 0.00\%$	$1.00 \pm 0.00\%$	$1.0\times$ (drive floor)

With mean weight pinned to the linear baseline, dispersion alone (σ from 0.5 to 1.6) leaves the ratio nearly unchanged ($17.1\times \rightarrow 14.5\times$); **dispersion is not the mechanism**. The compression to $1.75\times$ in the raw- w_{syn} anchor comes from *gain*: raising mean $|w| \sim 3.6\times$ pushes the female network past its ignition threshold ($2.8\% \rightarrow 45\%$) far more than the male ($40.9\% \rightarrow 79\%$). The honest summary is that the recruitment *ratio* depends on the operating point ($26\times$ threshold regime, $1.75\times$ near saturation), while the underlying sensitivity, how much less drive the male network needs for equal recruitment, is stable across rules (§3.5).

3.5 The branching-ratio estimator saturates; recruitment curves are shifted

The branching ratio prints 0.980–1.004 across all 130 unique ledger records, but at these drive densities that number carries no criticality information: with mean spikes per 1 ms bin $\lambda \geq 7.17$ in every condition, the mean-ratio estimator is bounded by the Poisson ceiling $1 - \exp(-\lambda) \geq 0.99923$, and the measured values sit on that ceiling (median deviation 0.0004; max 0.0197, at the saturated DES-male end). Randomizing spike times at fixed counts leaves the estimate unchanged ($\Delta \leq 0.0003$), and under sparse drive the same estimator separates regimes cleanly (0.057 dead vs 0.988 propagating). The "self-organized criticality" reading of such near-unit branching ratios is therefore a property of the drive protocol, not the networks; under quiescence-referenced avalanche statistics neither the real networks nor their nulls show a clean power law, and real and null are not distinguishable at these statistics. What survives is the recruitment story: the sex difference is *where on the recruitment curve* each sits, not the curve's shape:

Table 5. Recruitment ladder ($\sigma = 1.6$; single seed-pair per point except the equivalence cells (male 0.02, 0.05; female 0.06, 0.15) and the anchor ($w_{\text{syn}} = 0.1$; 3 seed-pairs)).

w_{syn}	Female % fired	Male % fired
0.01	n/a	1.96
0.02	1.73	29.13 ± 0.50
0.03	n/a	44.13
0.04	16.57	n/a
0.05	n/a	64.91 ± 0.06
0.06	28.08 ± 0.22	n/a
0.10	45.04 ± 0.49	79.03 ± 0.16
0.15	58.04 ± 0.48	n/a

The male reaches equivalent recruitment at $\sim 3\text{--}3.5\times$ lower w_{syn} . The first matched pair is recruitment-equivalent at a $3.0\times$ grid ratio (male 0.02 $\rightarrow 29.13 \pm 0.50\%$ vs female 0.06 $\rightarrow 28.08 \pm 0.22\%$, within 1.1 pp); the second (male 0.05 $\rightarrow 64.91 \pm 0.06\%$ vs female 0.15 $\rightarrow 58.04 \pm 0.48\%$) is $3.0\times$ on the grid but 6.9 pp short of equivalence, and interpolating the female ladder puts her at 64.91% near $w_{\text{syn}} \approx 0.18$, i.e. $\approx 3.5\times$. This w_{syn} -equivalence is the cleanest scale-free statement of the sex difference at the $\sigma = 1.6$ lognormal rule: $\sim 3\times$ on the first ladder pair, $\sim 3.5\times$ interpolated on the second.

3.6 Silencing dimorphic neurons: modest, distributed, null-controlled

Silencing all edges touching the 2.4% dimorphic neurons reduces male recruitment by 8.4% relative ($35.20 \rightarrow 32.23 \pm 0.45\%$). The size-matched random-ablation null (5 build seeds $\times$ 3 simulation seeds, $n = 15$) reduces it by 6.3% ($32.98 \pm 1.41\%$). The dimorphic-silenced recruitment sits 0.75 pp below the ablation null, but given the null's across-seed spread the two are statistically indistinguishable (Welch $t = 1.67$, $p \approx 0.12$); the dimorphic-specific effect is modest at best, and we cannot claim it exceeds random neuron loss. The sex difference survives silencing almost intact. In the female, silencing her dimorphic set is indistinguishable from random ablation (1.33 ± 0.29 vs 1.31 ± 0.23).

Table 6. Silencing with null (linear weights; mean $\pm$ SD; ablation null over 5 build $\times$ 3 simulation seeds).

Network	Real	Dimorphic-silenced	Random-ablated (null, $n = 15$)
Male MCNS	$35.20 \pm 0.68\%$	$32.23 \pm 0.45\%$ (-8.4%)	$32.98 \pm 1.41\%$ (-6.3%)
Female BANC	$1.35 \pm 0.29\%$	$1.33 \pm 0.29\%$ (-1.5%)	$1.31 \pm 0.23\%$ (-2.7%)

The sex difference is a distributed property of whole-network wiring, not a localized effect of annotated sex-specific circuits: dimorphic silencing does not measurably exceed the random-ablation null in either sex, consistent with the distributed dimorphism reported for the male CNS connectome (Berg et al., 2026).

3.7 Sign structure is net-suppressive in both sexes

Shuffling excitatory/inhibitory signs (preserving all wiring and sign counts) *increases* recruitment in both sexes: male $35.20 \rightarrow 61.12 \pm 2.88\%$, female $1.35 \rightarrow 13.16 \pm 10.50\%$ (Table 7). Real E/I placement is therefore actively suppressive relative to random sign placement; inhibition is not decorative. This also means DES rewiring (§3.3), which permutes edges and thereby remixes E/I composition, inherits a confound between wiring specificity and E/I arrangement; the §3.3 decomposition is bounded by this. The female's shuffled network is bimodal across seeds (19.3%, 19.2%, 1.0%): it sits at an ignition threshold where two of three seed draws escape the drive floor and one does not. We report the mean $\pm$ SD with the bimodality stated, and treat the female $\sigma = 0.5$ anchor ($3.98 \pm 2.07\%$, seeds 6.15/2.02/3.77) with the same candor.

Table 7. Sign-shuffle control (linear weights, $w_{\text{syn}} = 0.1$; mean $\pm$ SD over 3 seed-pairs).

Network	Real	Sign-shuffled
Male MCNS	35.20 $\pm$ 0.68%	61.12 $\pm$ 2.88%
Female BANC	1.35 $\pm$ 0.29%	13.16 $\pm$ 10.50% (bimodal: 19.3/19.2/1.0)

3.8 Controls for rate-confounding in higher-order metrics

When comparing networks with varying activity levels, metrics that scale with firing rate are highly susceptible to artifactual divergence: the more active network differs on such metrics for reasons unrelated to wiring (a caution for connectome-dynamics applications of such metrics generally; Suárez et al., 2024). We demonstrate this here by evaluating transfer entropy, participation ratio, and CV of ISI under per-neuron rate matching, in which the male's spike train is subsampled to the female's per-neuron rate before metric computation:

- **Transfer entropy.** The raw TE difference (0.0221 vs 0.0005, a "44 $\times$ " gap) collapses to 0.9 $\times$ under per-neuron rate matching (0.00047 vs 0.00051). The apparent information-flow advantage is a firing-rate artifact: under this manipulation (which thins only the male's train; one-sided thinning can itself impose or erase structure, as the PR case shows) the sexes are not distinguishable in information flow.
- **Participation ratio.** The raw PR suggests the male is lower-dimensional (1.28 vs 6.08, "more synchronized"). Under rate matching the male appears *more* high-dimensional (43.1 vs 6.1), but this reversal is itself an artifact: spike thinning decorrelates neurons and inflates PR for both sexes (the female's own PR rises from 6.08 to 19.5 when thinned to a common 2 Hz rate). Dimensionality estimated from post-hoc thinning is not defensible in either direction; a matched-rate simulation would be required for any claim.
- **CV of ISI.** The raw CV difference (2.99 vs 1.01) is rate-driven; a rate-matched CV comparison would require an archived matched-rate control, which we do not ship, so we make no rate-matched CV claim. Moreover, CV **reverses sign between operating regimes** (linear: male 2.74 vs female 1.04; $\sigma = 1.6$ anchor: female 2.68 vs male 1.48), so a single-regime CV comparison is uninterpretable without rate context.

4. Discussion

The finding. Embedded in an identical dynamical model, the male *Drosophila* CNS recruits substantially more of itself than the female CNS at matched drive. Under lognormal weights matched to the literature's fitted log-spread, the recruitment ratio is $1.75\times$ [1.74–1.78] with the male at 79% and the female at 45%; under linear weights it is $26\times$; under mean-matched lognormal it is $14.5\text{--}17\times$. The scale-free statement: the male reaches equal recruitment at $\sim 3\text{--}3.5\times$ lower synaptic weight scale ($3.0\times$ on the first matched ladder pair, $\approx 3.5\times$ interpolated on the second; 3-seed).

The decomposition. At the saturated anchor, degree-preserving rewiring splits the log-gap roughly 58/42 between specific wiring and degree/density structure: two-fifths (42%) of the male's recruitment advantage is available from degree structure alone (more edges per neuron), and the majority from which specific edges exist. Equally important, both networks' specific wiring reads as net-suppressive relative to their own degree skeletons there, the female's far more so (0.47 vs 0.15 log units). Three bounds temper this decomposition. The split is scale- and seed-limited (57.7% [56.9, 58.5] bootstrapped on the log-ratio scale; 43/57 on the linear scale; §3.3). It is operating-point-dependent, not a general property: re-simulated across w_{syn} , the specific-wiring share runs from -260% near threshold to $+58\%$ at the anchor (Table 3b). And DES rewiring permutes edges but thereby also remixes each network's E/I composition, which is itself suppressively organized (§3.7); wiring specificity and E/I arrangement are confounded in this null. Within those bounds, the female connectome's damping remains the anchor-level story.

Criticality. The near-unit branching ratios across conditions are a property of the drive protocol, not evidence of criticality: at these drive densities the mean-ratio estimator is pinned to its Poisson ceiling $1 - \exp(-\lambda)$ regardless of regime (§3.5), and under quiescence-referenced statistics the real networks and their nulls are not distinguishable. We therefore make no criticality claim; at the operating points studied the networks are continuously active and near-saturating. The sex difference manifests as position on a shared recruitment curve, reached at lower coupling in the male. Reports of self-organized criticality in the fly connectome (Shiu et al., 2024) motivate the question, but the mean-ratio estimator at this drive protocol cannot adjudicate it; a quiescence-referenced protocol (§3.5) would be required.

Why the magnitude question is regime-relative. The recruitment ratio depends on where the networks sit relative to their ignition thresholds: near-threshold regimes amplify the gap ($14\text{--}27\times$); high-gain regimes compress it ($1.75\times$). Reporting a single ratio without its operating point obscures the underlying network dynamics. We recommend connectome-dynamics

comparisons report (i) the recruitment ladder (§3.5), (ii) the w_{syn} -equivalence ratio, and (iii) any operating-point ratio with its regime stated, not a headline number alone.

Methodological contributions. (1) The dual-seed SSOT protocol (independent weight and dynamics seeds, append-only ledger, bootstrap CIs on ratios). (2) The DES decomposition as a routine control for connectome dynamics: it is cheap, degree-exact, and separates wiring effects from degree/density effects. (3) The ablation-null standard for silencing claims. (4) The rate-matching controls of §3.8 as a demonstration of how higher-order metrics fail across activity regimes.

5. Limitations

- **Synaptic weights are inferred, not measured.** Counts are a proxy; the lognormal rule is synthetic (i.i.d. log-noise on counts), not a per-edge effectome fit (Pospisil et al., 2024). Our spread-matching ($\sigma = 1.6 \Rightarrow \text{SD}(\log|w|) \approx 1.83$) matches the literature's marginal log-spread but not its structure.
- **Operating-point dependence.** The $1.75\times$ headline holds at the high-recruitment anchor; threshold-regime ratios are $14\text{--}27\times$. We consider the w_{syn} -equivalence ($\sim 3\text{--}3.5\times$) the most transferable quantity.
- **Ablation-null spread.** Nulls are simulated under all three dynamics seeds (DES 3 build $\times 3$; ablation 5×3). DES cross-seed SD is small (within-build ≤ 0.066 pp), but the ablation nulls spread widely (within-build SD up to 2.65 pp male, 0.28 pp female), and the dimorphic-silencing contrast is not statistically resolved at current seed counts (§3.6); resolving it would require more seeds.
- **Ladder points at $w_{\text{syn}} = 0.01, 0.03, 0.04$, and the female 0.02 cell are single-seed;** the four equivalence cells (male 0.02, 0.05; female 0.06, 0.15) and the anchor are 3-seed.
- **No gap junctions, neuromodulation, or conduction delays.** Known dynamics-shaping omissions.
- **Amine signs are a fixed ionotropic convention.** Dopamine, octopamine, tyramine (+1) and serotonin (−1) act largely through metabotropic receptors in *Drosophila*; the sign-shuffle result (§3.7) and the DES E/I-composition confound (§3.3) are conditional on this coarse map.
- **Dimorphism annotation is incomplete;** the 2.4% (male) / 2.2% (female) figures are lower bounds, and dimorphic-set silencing inherits annotation bias. Berg et al. (2026) report a larger dimorphic fraction ($\sim 5\%$) for the male CNS; our annotation-flag sets are narrower by construction, which strengthens rather than weakens the distributed-storage reading of §3.6.

- **Transfer entropy used simplified binned estimation;** a proper estimator (e.g., JIDT) would be required for any use of that metric.
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6. Conclusion

The male and female *Drosophila* CNS connectomes, embedded in an identical LIF model, differ robustly in intrinsic recruitment: the male recruits $1.75\times$ [1.74–1.78] more of its network under literature-matched lognormal weights, and reaches equal recruitment at $\sim 3\text{--}3.5\times$ lower synaptic weight scale under the $\sigma = 1.6$ lognormal ladder tested. Degree-preserving decomposition attributes $\sim 58\%$ of the log-scale difference to specific wiring and $\sim 42\%$ to degree/density structure at the saturated anchor, with both connectomes' specific wiring reading as net-suppressive there; the split varies with operating point (Table 3b) and inherits an E/I-composition confound (§3.7). The near-unit branching ratios reflect the drive protocol rather than criticality; the sex difference is position on a shared recruitment curve. Dimorphic-circuit silencing and its size-matched null show the difference is distributed, not localized: silencing does not measurably exceed random ablation in either sex. Rate-matching controls demonstrate that transfer entropy and participation ratio are vulnerable to artifactual divergence across activity regimes and should not be used as headline metrics in such comparisons. We release the dual-seed SSOT protocol, the degree-preserving DES null, and the full results ledger as reusable standards for connectome-dynamics comparison.

References

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Reproducibility

All numbers in this paper derive from append-only ledgers: results/ssot/ssot_results.jsonl (130 unique records, 133 lines, dual-seed; aggregated by aggregate_ssot.py into results/ssot_synthesis.md) and companion probe ledgers under results/crit_probe/ (Table 3b; §3.5 regime tests), with archived spike-train artifacts backing per-metric statistics (§3.8) and connectome edge lists backing the structural statistics of the §2.1 data table.

Pipeline:

1. normalize.py: connectomes $\rightarrow$ common edge list
2. double_edge_swap.py: degree-preserving DES nulls ($Q = 10$, 3 seeds $\times$ 2 sexes, degree-verified)
3. ablation_null.py: size-matched random ablation nulls (5 seeds $\times$ 2 sexes)
4. null_model.py: sign-shuffle control
5. ssot_run.py: the dual-seed LIF experiment (weight_seed for weight draws, sim_seed for dynamics + drive selection; brian2.seed() explicit)
6. gen_queue.py: condition enumeration (46 (tag, sex) condition families; see Results)
7. aggregate_ssot.py: dedupe (last-timestamp-wins), mean $\pm$ SD, bootstrap ratio CIs $\rightarrow$ ssot_synthesis.md

Every result record carries

{tag, source, transform, sigma, w_syn, weight_seed, sim_seed, n_neurons, n_edges, pct_fired, rate_all_Hz, branching_ratio, cv_isi, ...}. Re-running the queue reproduces the ledger modulo wall-clock.